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Problem 1. Principle Component Analysis

A) Visualize the raw data for all of the points in the 3rd versus the 18th dimensions in the datapoints.csv file.

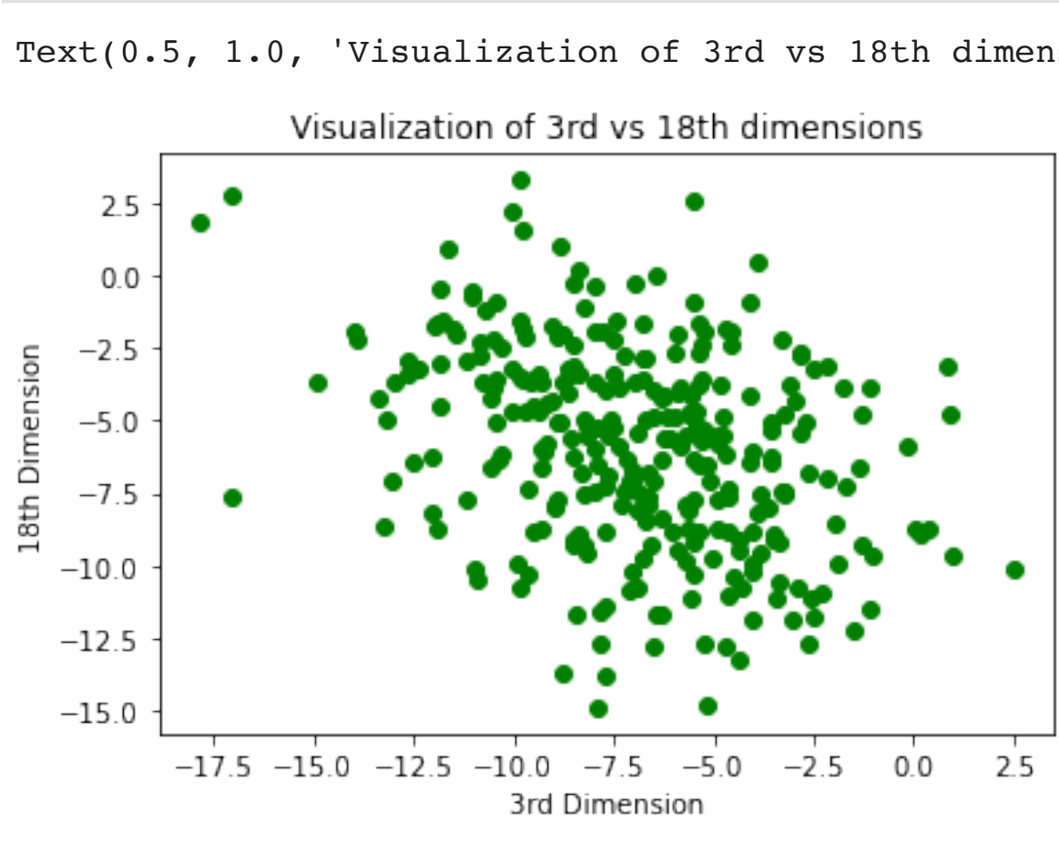
In [1]:
```
# do basic imports
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as sps
import pandas as pd
import pickle
```

In [2]:
```
data = pd.read_csv('datapoints.csv') #import data
df = pd.DataFrame(data) #
#print(df.head())
print(df.shape)
print(df['0'].shape)
```

```
(300, 20)
(300,)
```

In [3]:
```
plt.plot(df['3'],df['18'],'o', color = 'green')
plt.xlabel('3rd Dimension')
plt.ylabel('18th Dimension')
plt.title('Visualization of 3rd vs 18th dimensions')
```

Out[3]:
```
Text(0.5, 1.0, 'Visualization of 3rd vs 18th dimensions')
```



B) Perform a principle component analysis (PCA) on the data and return all the principle components. Show the principle components in a pandas data-frame.

In [4]:
```
from sklearn.decomposition import PCA
```

In [5]:
```
#finding the principle components
pca = PCA(n_components = 20) #this defines the model, there are 20 columns so that means 20 variables so 20 PC components
p = pca.fit_transform(df)
pcadf = pd.DataFrame(data = p, columns = ['PC1', 'PC2', 'PC3', 'PC4', 'PC5', 'PC6','PC7','PC8','PC9', 'PC10','PC11','PC12','PC13','PC14','PC15','PC
print(pcadf)
```

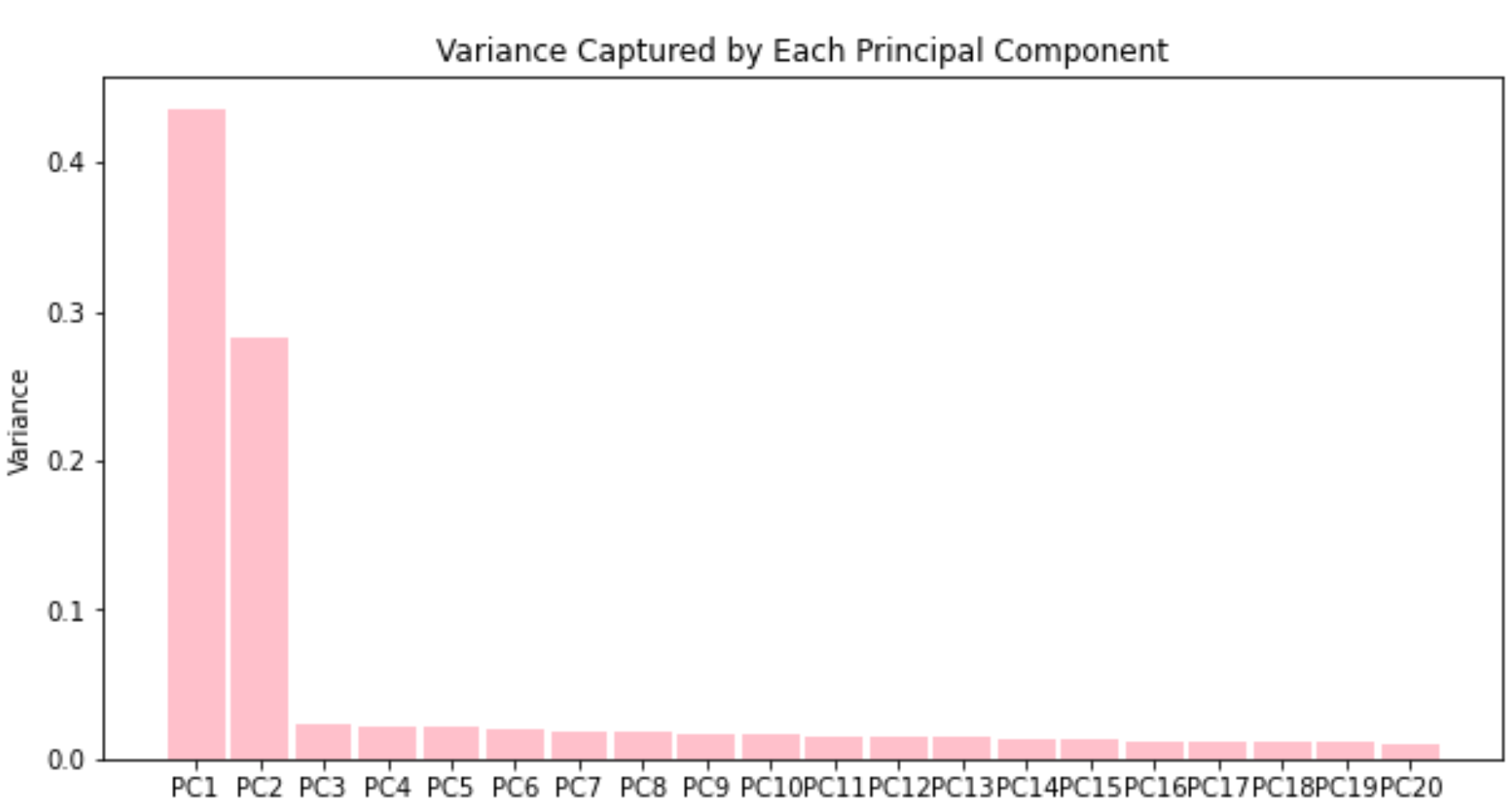
```
           PC1        PC2        PC3       PC4       PC5       PC6       PC7  \
0    19.449531  -9.205561   1.483845  0.298622  1.907189  1.932403  1.805941
1    14.256032  -4.787531   0.635671  4.235459 -0.663979  8.090297 -2.070789
2     2.614282  16.753922  -0.460214  3.850638 -1.667367 -2.516891 -5.088551
3   -13.155587  -4.439978   2.577246 -4.931699 -0.688698  3.164437  1.635518
4   -15.915419  -5.929977   2.415124  1.652355  0.199788 -2.590344 -1.357000
..         ...        ...        ...       ...       ...       ...       ...
295   0.019962  16.350879   5.228203  0.194244  5.513870  3.494995  2.170749
296  -3.050148  21.867717  -1.812844 -1.445882  0.748240  0.859756 -0.486510
297 -15.168477  -6.608786   3.552535  2.486504  1.815216 -4.469368 -1.949865
298   1.322555  22.126099  -5.463404  0.061418 -2.887073  1.963890  1.702409
299   1.473089  12.065865  -1.427828 -7.251994 -2.492307  0.538678  4.654171

          PC8       PC9      PC10      PC11      PC12      PC13      PC14  \
0   -2.679595  2.446757 -2.075839  0.872516 -0.367090  2.786740  0.060869
1   -1.095881  1.126791  0.688756  2.959866  0.253803 -2.781194  1.821687
2    0.924928 -1.465603 -3.362423  3.340204  0.883720  2.255539  2.753372
3    1.731904 -0.785951 -1.560959 -1.833057 -5.273586 -1.408587  0.092609
4    0.463776  1.204803  0.897765 -2.079389 -4.176975 -2.678965  3.890564
..        ...       ...       ...       ...       ...       ...       ...
295 -4.777813 -5.036545 -7.685108  2.326386 -0.208037  2.373163 -1.235100
296 -0.632137  0.182717 -1.894684 -1.752805  0.414478 -2.149127 -0.657049
297  0.491122  4.391598  2.215201 -3.178532  3.596151 -0.206664  2.438704
298  1.471243  1.781278  0.500924 -5.090575 -3.772271 -0.128906 -4.011306
299  2.777586 -0.949444  2.074630 -1.765247 -1.223576  0.008477 -0.123500

         PC15      PC16      PC17      PC18      PC19      PC20
0    1.176195  0.462079  1.679361 -1.645234  1.295988  3.165279
1    0.248842  1.805805  1.870599  3.794675 -0.706354  2.495231
2   -2.288532  0.995473 -0.640504 -5.210043  1.803922  0.279415
3    1.507077 -0.877069 -0.499607 -1.232077  2.321160  0.098330
4    3.941198  0.217374 -0.122535 -1.982663 -2.199513 -2.684517
..        ...       ...       ...       ...       ...       ...
295  2.796031 -0.869893 -0.422509 -1.754012 -0.252129  5.154510
296  1.132381 -2.597553 -1.900000  2.046820  0.063959  1.903590
297  3.335932  1.698664  1.923570 -2.532367 -0.399942 -0.080272
298  0.460597 -0.337045 -5.510235  0.244777 -0.843082 -0.341277
299  4.302659 -2.883582 -4.934991  5.350930 -1.084200 -0.673795

[300 rows x 20 columns]
```

C) Plot the variance captured by each principal component. Based on this plot, ex- plain if PCA is a good option to reduce the dimensions of the given data.

In [6]:
```
#make a plot where y-axis is variance and x-axis is the principle component
v = pca.explained_variance_ratio_ #variance captured for each principal component
print(v)
#varper = np.cumsum(np.round(v,decimals=3)*100) #percent variance explained for each principal component
#print(varper)
pc = pcadf.keys() #get the PC names for the x-axis
print(pc)
fig = plt.figure(figsize = (10,5)) #increase the figure size
plt.bar(pc,v, color = 'pink', width = 0.9)
plt.xlabel('Principal Component')
plt.ylabel('Variance')
plt.title('Variance Captured by Each Principal Component')
```

```
[0.43564073 0.28191603 0.02329473 0.02145605 0.02091404 0.0206554
 0.01796797 0.01758981 0.01711461 0.01572273 0.0154126  0.01461655
 0.01417692 0.01360172 0.01300943 0.01224864 0.0121201  0.01145212
 0.01099109 0.01009872]
Index(['PC1', 'PC2', 'PC3', 'PC4', 'PC5', 'PC6', 'PC7', 'PC8', 'PC9', 'PC10',
       'PC11', 'PC12', 'PC13', 'PC14', 'PC15', 'PC16', 'PC17', 'PC18', 'PC19',
       'PC20'],
      dtype='object')
```

Out[6]:
```
Text(0.5, 1.0, 'Variance Captured by Each Principal Component')
```



Explain if PCA is a good option to reduce the dimensions of the given data based on the plot

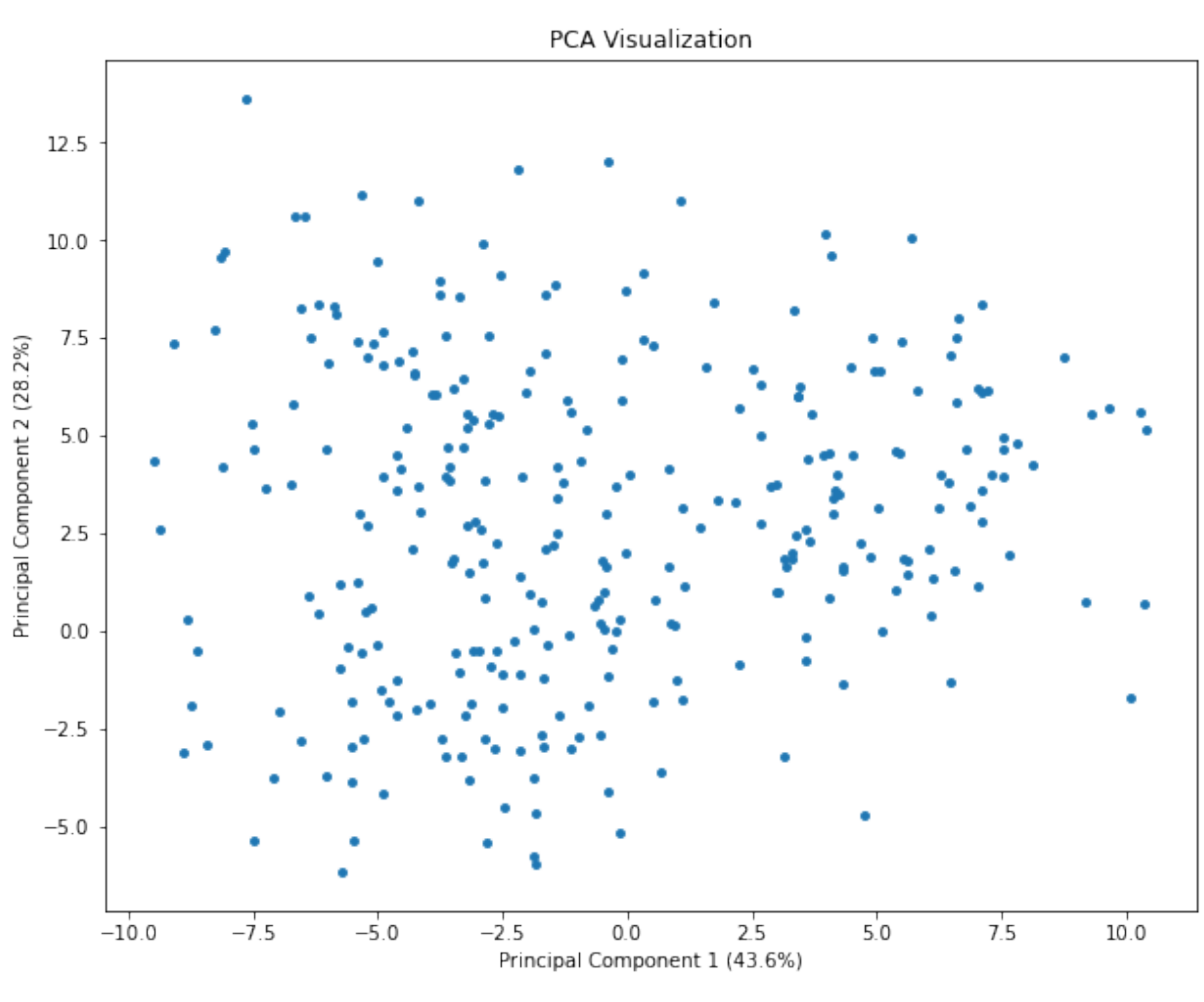
I believe yes because most of the variance is explained by the first two PC in the plot and the rest of the principal components do not explain that much variance.

D) Plot the first two principal components. Make sure to include the variance cap- tured by the first two principal components in the axis labels.

In [7]:
```
X_pca = pca.transform(p) #transform the data into PCA space
print(len(p[:,0])) #that is principal component 1
print(len(p[:,1])) #that is principal component 2
#x = p.transform(df)
#target = np.zeros(df.shape[0])
plt.figure(figsize=(10,8))
plt.scatter(X_pca[:,0],X_pca[:,1], s=15)
plt.xlabel('Principal Component 1 (43.6%)')
plt.ylabel('Principal Component 2 (28.2%)')
plt.title('PCA Visualization')
#need to add variance captured by the first two principal components in the axis labels
```

```
300
300
```

Out[7]:
```
Text(0.5, 1.0, 'PCA Visualization')
```



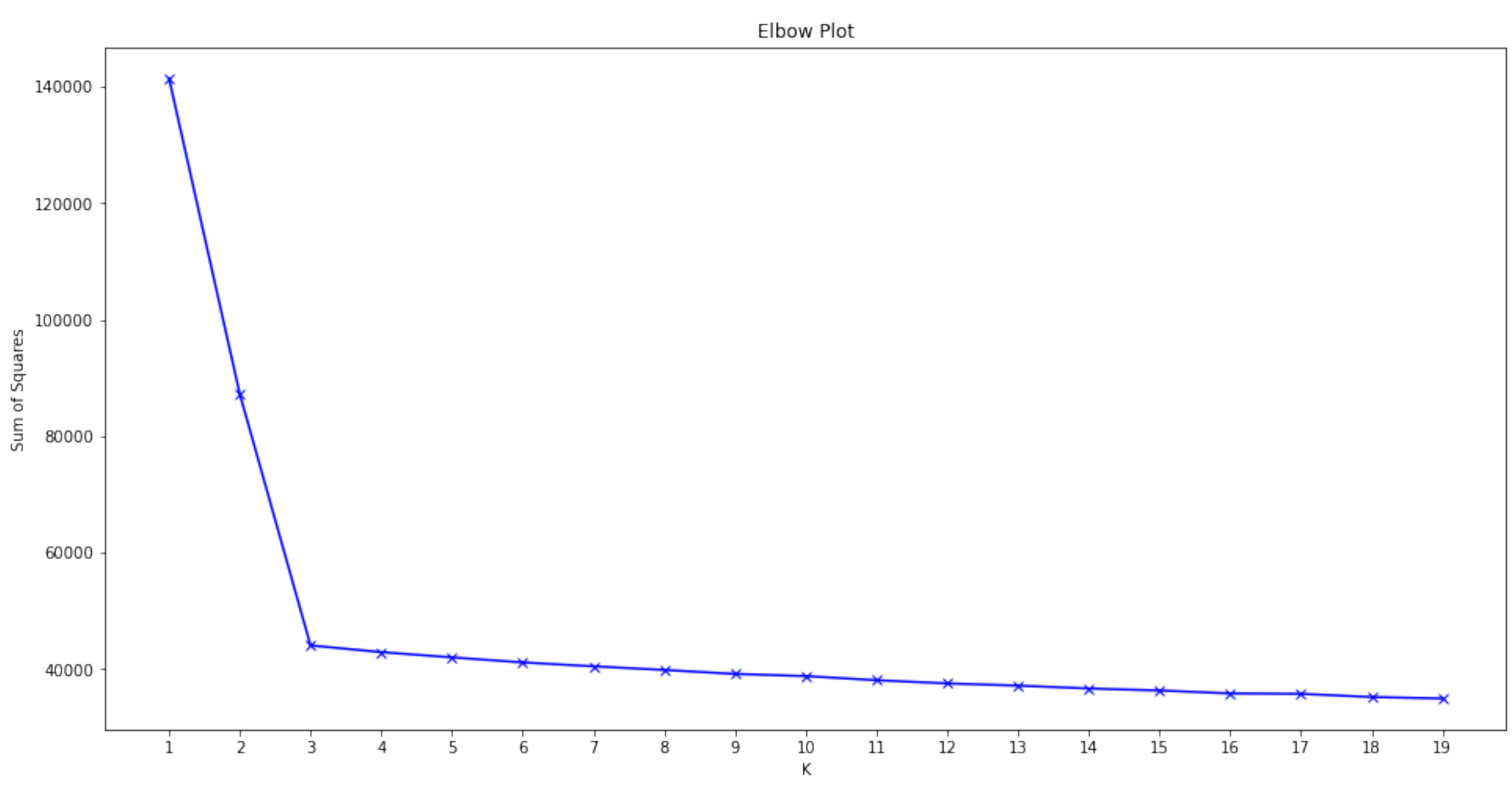
Problem 2. K-Means Clustering

A) Using the principle component data generated from problem 1A, plot the sum of squared distances of all points in a cluster for all principal components (1-20). Please use sklearn's KMeans method and its attribute kmeans.inertia to obtain the sum of squared distances. From this plot, you should be able to deduce the optimal number of clusters. Please report the optimal number of clusters based on your plot.

In [8]:
```
#plot the sum of squared distances of all points in a cluster for all principal components
#used sklearns Kmeans methods and kmean.intertia to obtain the sum of squareed distances
```

In [9]:
```
from sklearn.cluster import KMeans
inertia = [] #empty list in order to place the values from the inertia
 #there are 20 prinicipal components so make it go through a range of 20
point = range(1,20)
for i in range(1,20):
    kmean = KMeans(n_clusters = i)
    kmean.fit(pcadf)
    inertia.append(kmean.inertia_)

plt.figure(figsize=(16,8))
plt.plot(point,inertia, 'bx-')
plt.xticks(point) #set up the tick marks correctly
plt.xlabel('K')
plt.ylabel('Sum of Squares')
plt.title('Elbow Plot')
```

Out[9]:
```
Text(0.5, 1.0, 'Elbow Plot')
```



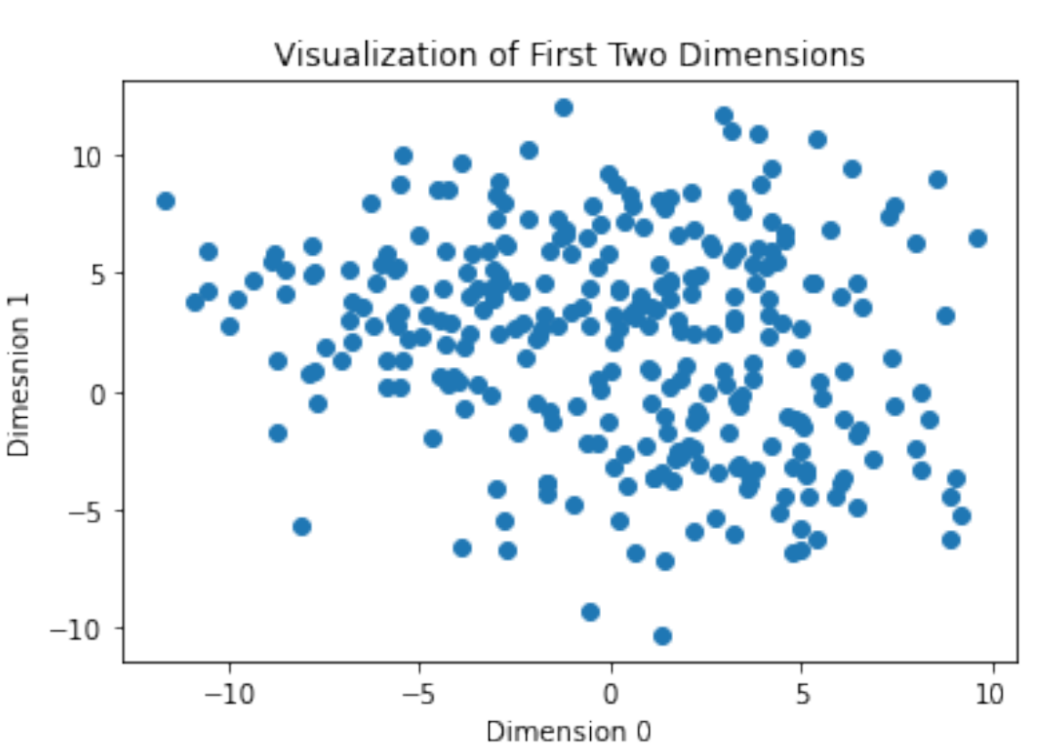
In [10]:
```
#select value of K at the elbow of the graph which is the point after the interia started decreasing
#based on the plot I would say the optimal number is k=3
```

B) Perform K-Means clustering using the first two dimensions of the ORIGINAL DATA (data from the datapoints.csv). To determine the number of clusters that you should begin with, refer to your elbow plot from problem 2A, and randomly generate the points of centroids according to that cluster number (i.e. if you reported the optimal number of clusters to be 10 in problem 2A, you should start with 10 centroids). Before generating these centroids, set your random seed as follows: random.seed(10). Perform 10 iterations of K-Means. Finally, plot the K-Means clusters. It should be clear on the plot as to what cluster each point belongs to (i.e. each cluster should be a different color). This plot should also contain the final calculated centroid for each cluster. Make sure to include a legend for your plot.

In [11]:
```
#make multiple functions
#one to calculate the distance betweeen poitns and the centroids
#one to calculate a list of points close to each centroids
#one to calculate the new centroid from the list of points in that cluster

zero = df['0'].values #subset the orginal dataframe so that we are only looking at the first two dimensions of th eorginal data
one = df['1'].values
partb = np.array(list(zip(zero,one))) #combine the two values together

#iterations = 10 #amount of times we are going to run the simulations
#K = 3 #number of clusters
```

In [12]:
```
plt.scatter(zero,one)
plt.xlabel('Dimension 0')
plt.ylabel('Dimesnion 1')
plt.title('Visualization of First Two Dimensions')
```

Out[12]:
```
Text(0.5, 1.0, 'Visualization of First Two Dimensions')
```

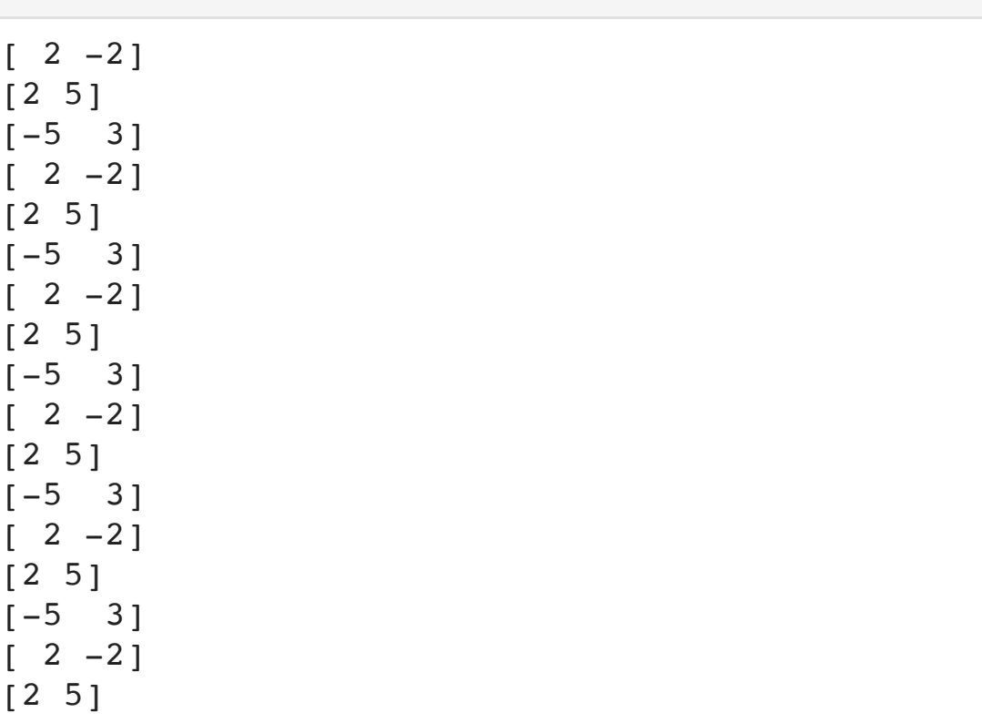


In [18]:
```
#first function
import random
#calculating the distance between points and the centroids
def randomcentroids(Y,k):
    '''Obtaining the random centroids from the data '''
    #for i in range(k):
    random.seed(3)
    #using random random int into a value that is betweeen 0 and the max value of the dataset
    coord1 = np.random.randint(0,np.max(Y),size= k) #getting the random random value for x axis and getting three if them because three centroids
    coord2 = np.random.randint(0,np.max(Y),size =k) #getting the random random values for the y-axis
    centcord = np.array(list(zip(coord1, coord2))) #putting them together in order to get x and y coordinates
    return centcord
c = randomcentroids(partb, 3)
print(c)
#print(c[0])
#print(c[0,0],c[0,1])
#print(c[1,0],c[1,1])
#print(c[2,0],c[2,1])
```

```
[[ 0  1]
 [ 8  8]
 [ 1 10]]
```

In [19]:
```
plt.scatter(zero,one)
#plotting the random centoids that were made to see how it is in the data
#using a star to mark the centroid
plt.scatter(c[0,0],c[0,1], marker = "*", s =150) #adding the centroid points into the plot to visulize them
plt.scatter(c[1,0],c[1,1] ,marker = "*", s = 150)
plt.scatter(c[2,0],c[2,1] , marker = "*", s= 150)
```

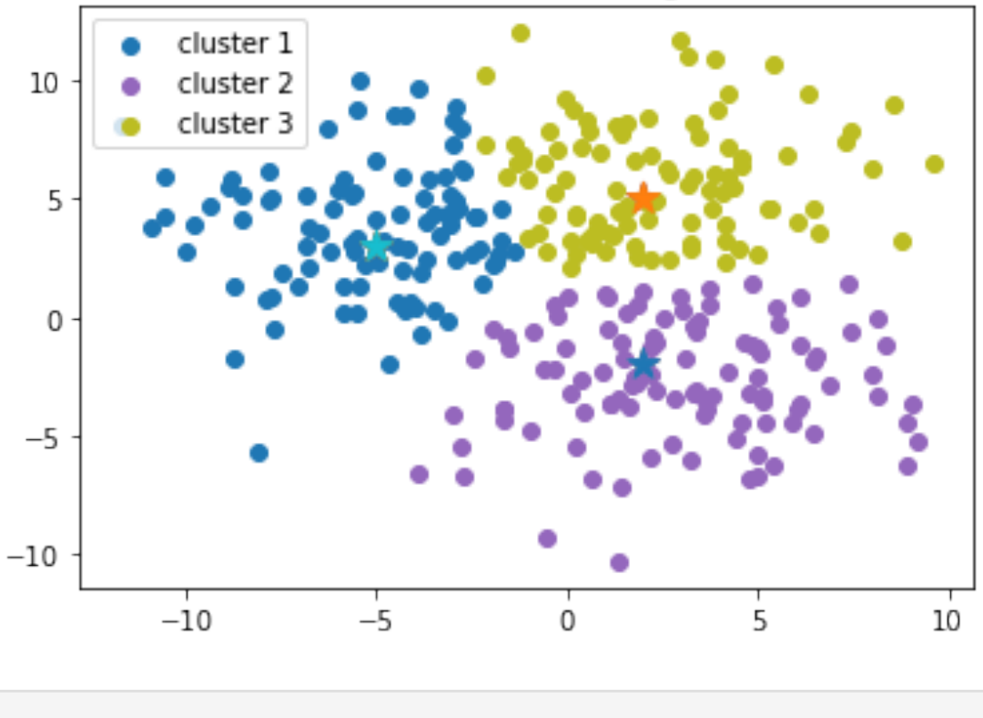
Out[19]:
```
<matplotlib.collections.PathCollection at 0x7fea708d8a30>
```



In [22]:
```
def distance(Y,k,c):#function to make the distances and it will take in the data, number of clusters, and centroids made before
    '''taking the random clusters formed previously and assinging each point to a cluster then calculating means again to get the clusters, process
    center = []
    for i in range(10): #we are running the iteration ten times
        dim= Y.shape[0] #get the shape of the data
        idx = np.zeros(dim)
        for i in range(dim):
            clus = np.zeros(k) #make a numpy array
            for j in range(k):
                clus[j] = np.sum((Y[i,:]-c[j,:])**2) #calculating the distance between the points and the centroid
                idx[i]= np.argmin(clus) #take the arg min
        for l in range(k):
            points = [Y[j] for j in range(len(Y)) if idx[j] == l]
            c[l] = np.mean(points, axis=0) #calculating the means based on the points in order to get the new centroids
            print(c[l]) #printing all the new clusters just to see
    return idx
d= distance(partb,3,c)
#print(d)
#print(center)
```

```
[ 2 -2]
[2 5]
[-5  3]
[ 2 -2]
[2 5]
[-5  3]
[ 2 -2]
[2 5]
[-5  3]
[ 2 -2]
[2 5]
[-5  3]
[ 2 -2]
[2 5]
[-5  3]
[ 2 -2]
[2 5]
[-5  3]
[ 2 -2]
[2 5]
[-5  3]
[ 2 -2]
[2 5]
[-5  3]
[ 2 -2]
[2 5]
[-5  3]
[ 2 -2]
[2 5]
[-5  3]
```

In [17]:
```
cluster = ['cluster 1', 'cluster 2', 'cluster 3']
for i in range(3):
    points = np.array([partb[j] for j in range(len(partb)) if d[j]==i])
    plt.scatter(points[:,0], points[:,1], label = cluster[i])
    plt.legend()
    plt.scatter(-5,3,marker = "*", s=150)
    plt.scatter(2,-2,marker = "*", s = 150)
    plt.scatter(2,5, marker = "*", s = 150)
    plt.title("K-means Clustering")
```



In [ ]: